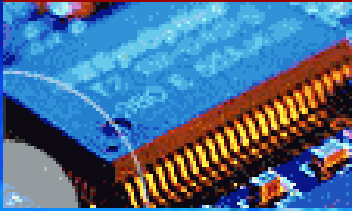
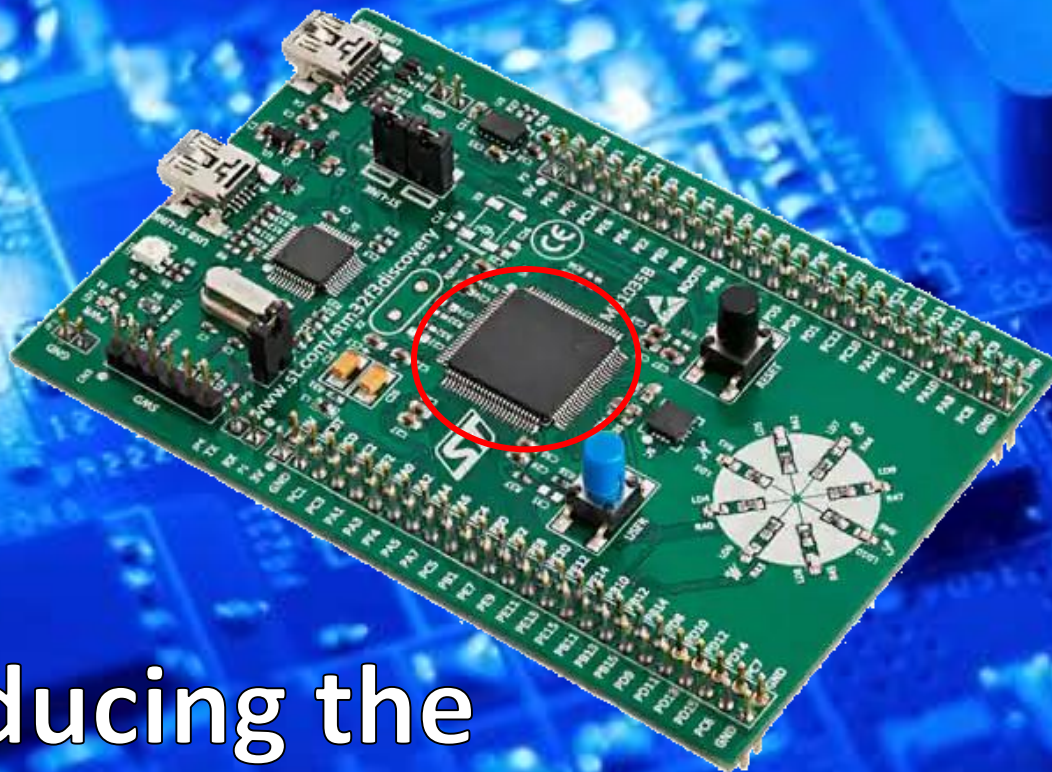


EEE3096S



**Embedded
Systems II**



ARM

Advanced RISC Machines

Part1/2

Introducing the Advanced RISC Machine (ARM) Embedded Systems II

Lecture L06

Dr Simon Winberg



Electrical Engineering
University of Cape Town

Outline of Lecture

- A Brief History of ARM: lead-up to ARM
- RISC vs. CISC
- (Powwow moment: RISC vs. CISC)
- Alternatives to ARM
- ARM cores & their application areas
- Where ARM microprocessors are used

Condensed slideshow to set the scene for ARM assembly programming and to give brief awareness of ARM The Business and ARM The Processor.

The ARM logo consists of the letters "ARM" in a bold, blue, sans-serif font. A small registered trademark symbol (®) is located at the top right of the letter "M".

Advanced RISC Machines

Powwow* concept

Consult the four winds, and your neighbouring classmates, as to:

Brainstorm an idea or solution

HINT: For such a thing there is probably various decisions on the purpose and focus etc. before the grant start. So we can simplify and have the 'elder' reflect on some past history, tell some stories to get the community thinking before deciding and doing.

We will have a powwow shortly ...



* It's a term from North America's Native people used to refer to a cultural gathering.

The origins of ARM....

Acorn Computers



- Started in 1978 as a subsidiary of Cambridge Processing Unit, Ltd.
- The company's first project was a slot machine, developed on a tiny budget.
- Very successful in 1980s and early 1990s
- Bought by Olivetti in 1985.

Olivetti Sowden OFX
Executive Fax machine



Olivetti Today: Italiano corporation that specializes in very expensive (and good looking) products like most Italian companies, but instead of manufacturing genuine leather shoes, they develop office equipment for the European market, especially very highly priced printers, faxes and multi-purpose printer/fax/scanner machines.

See [/www.olivetti.com](http://www.olivetti.com) In 2008 Olivetti celebrated it's centenary.



Acorn Computers BBC Micro



BBC Micro's processor:
Motorola 65002



Acorn's BBC Micro, from 1982, was one of the best home computers of that decade.

ARM History



- 1983: Acorn begin design of a processor - the Acorn RISC* Machine
- 1985: ARM1 launched, fabricated by VLSI Technology
- 1987: ARM processor debuts in a commercial product
- 1990: ARM Ltd spun off from Acorn, support from Apple
- 1991: ARM6 launch – ARM's first embeddable RISC core

ARM = **A**dvanced **R**ISC **M**achine



+

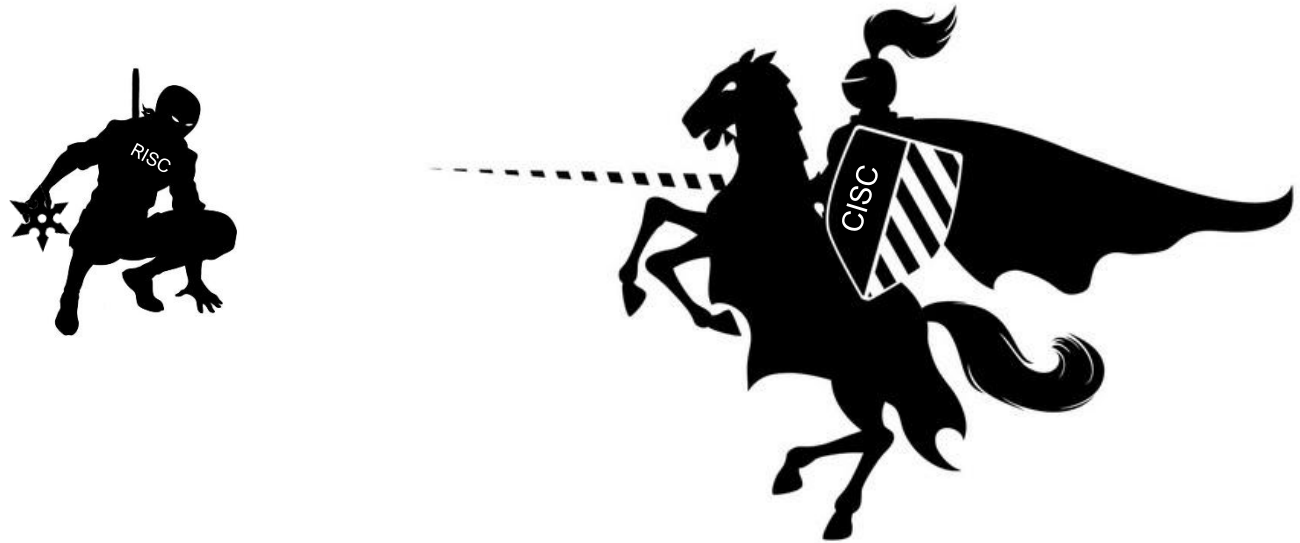


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* RISC = Reduced Instruction Set Machine

RISC vs. CISC



RISC = Reduced instruction set computer

CISC = Complex instruction set computer

Main aspects:

- RISC has less instructions, therefore needs more software
- CISC has more instructions, therefore needs less software
- Majority of today's microprocessors are RISCs

Characteristics of CISC

- Large number of instructions (100 – 250 instructions)
- Some **instructions that perform specialized tasks** and are used only infrequently
- Many **different addressing modes** (say 5 or more)
- Variable-length instruction formats
- Possibly **variable length instruction** execution cycles (e.g., ADD taking more cycles than LOAD)
- Instructions that **manipulate operands directly** in memory (e.g. add value at address X to reg. A)

Characteristics of RISC

- Small number of instructions (usually <100)
- Just **enough types of instructions** (or close to minimal)
- Few addressing modes (maybe be just two or three)
- Fixed-length instruction formats, easy to decode
- All **instructions take the same number of clock cycles** to complete (typically the case)
- All operations (e.g., ADD, COMPARE, etc) **work only with registers** (no memory access)
- Often use micro-programmed control (i.e., CPU instructions a set of even lower-level instructions)
- Memory access limited to LOAD and STORE instructions (i.e. no instructions manipulate operands in memory)

time for...

POWWOW

Powwow* moment

Thinking Point Posed during lecture:

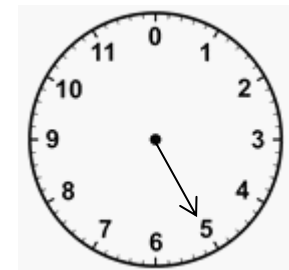
1 min get into groups / stretch ...

Discuss & jot down some points: 5 min. time limit!



Next slide provides some solutions...

**TIME
UP**



**Time
limit**

* It's a term from North America's Native people used to refer to a cultural gathering.

(answers will be added here)

Awareness of alternatives to ARM

Thinking about processor options

- You are aware of ARM ... (e.g. flavours by STM, Texas Instruments, NXP/Freescale, Intel...)
- An ES developer should also be aware of these often-used microcontrollers:
 - **PIC** (Peripheral Interface Controller)
 - **AVR** (or 'Alf and Vegard's RISC processor', an Atmel)

PIC and AVR are essential for designing low-power, low-cost systems!



Key Features of PIC

- Developed by Microchip Technology
 - (Derived from PIC1650 originally designed by General Instrument's Microelectronics Division)
- Available since 1976*
- Some use pure **Harvard Architecture**, i.e. program memory protected; others **Modified Harvard** where the program memory can be read (and sometimes written)
- All models use flash memory for program storage
- Low cost, low power, ease of reprogramming with built-in EEPROM
- Abundant development tools and application notes
- Often marketed as “PICmicro”
- A long history of use, probably the most used microcontroller of all time!



* Information source: https://en.wikipedia.org/wiki/PIC_microcontrollers

Key Features of AVR

- AVR – “Alf and Vegard's RISC processor”
 - Modified Harvard architecture 8-bit RISC
 - Even lower power options, generally low cost (maybe the cheapest AVR is not as cheap as the cheapest PIC)
 - ATmega328P one of the most popular, very low power but fairly powerful micros
 - Major claim to fame: one of the first micros to have on-chip flash, instead of one-time programmable ROM or hassles of using EEPROM
- Inspiring historical fact:
 - The AVR architecture was conceived by two students from the Norwegian Institute of Technology (NTH), Alf-Egil Bogen and Vegard Wollan*
 - The name “AVR” is commonly thought to stand for “Alf and Vegard's RISC processor”

Thinking Point: to ARM or not to ARM

Which of these applications do you think an ARM processor would be suited to?

- Coffee percolator
- ABS brakes
- Radio control of a model airplane
- Car radio (e.g., auto tuning, stored stations)
- Washing machine
- Portable battery-operated game console
- Weather sensor
- Garage door opener
- iPod (type of thing)
- Burglar alarm
- Radar car speed sensor
- Wireless network switch



... and which of these applications might you instead use an Intel 8051, AVR or PIC? Or even some other option?

Suggested for test practice ... more about explaining 'why' than choosing which.

Think Point Solutions:

Suggested choices & why

I'd probably make the following choices...

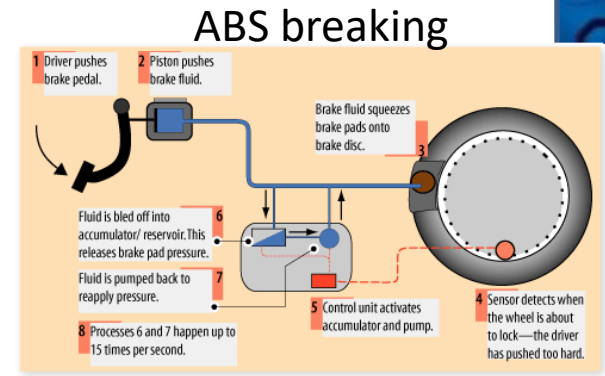
- Coffee percolator - PIC (:: simple, not much processing)
- ABS brakes - ARM or AVR (:: high-speed, not simple, fair amt processing)
- Radio control of a model airplane – PIC or AVR (:: simple, little processing)
- Car radio – PIC for auto turning feature at least
- Washing machine – Probably ARM, maybe PIC for very basic version
- Portable battery-operated game console - ARM
- Weather sensor – PIC (lowest power options, usually doesn't need speed)
- Garage door opener – PIC or AVR, something cheap
- iPod (type of thing) – ARM... although new iPod has Apple A10 SoC 4x 64bit cores
- Burglar alarm - probably a PIC, if there's not much signal processing
- Radar car speed sensor – probably a DSP and an ARM or PIC for comms
- Wireless network switch – perhaps a ARM or something super fast, like DSP

Which of these applications might you instead use an Intel 8051, AVR or PIC? Or even some other option?

Where are ARM cores used?

In a wide range of embedded computers and control systems e.g.

- Highly reliable – ABS, hard drive
- Consumer – cameras, PDAs smartphones (e.g. Nokia N series, Sony K series)
- Game systems (Nintendo DS, etc)
- Network systems (routers, switches, firewalls, etc.)



Cellphone



iPod



Nintendo DS



Network switch

Companies that use ARM

Just a some of the companies that use ARM cores:

Agere, AWS, Broadcom, Google, Fujitsu, Infineon, Intel, NEC, Nokia, Philips, Qualcomm, Texas Instruments, Toshiba, Sega, Sony, Nintendo, Cirrus Logic, Motorola, Thomson Multimedia .. vary many!

ARM licenses effectively used by all the top semiconductor companies.

A photograph of a closed red theater curtain. The curtain is a deep red color with vertical pleats and hangs from a dark metal track at the top. The lighting is dim, with a slight spotlight effect on the center of the curtain. At the bottom of the image, there is a dark red rectangular box with a thin blue border containing the word "intermission" in white text.

intermission